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DTCMamba: Dual-Branch Token-Chunking Mamba for Hyperspectral Image Classification

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***Abstract*—State Space Models (SSMs) are emerging as an efficient alternative to Transformers for hyperspectral image (HSI) classification, offering linear-time long-range modeling. However, existing Mamba-style pipelines often treat spectra as a single homogeneous sequence, amplifying redundancy and exacerbating long-range state decay; meanwhile, spatial flattening disrupts 2-D locality and induces boundary artifacts. We propose DTCMamba, a dual-branch token–chunking Mamba framework for spectral–spatial HSI classification. A history-augmented token–chunking SSM block injects gated second-order states to stabilize propagation over hundreds of bands, while chunk-level parallelism reduces sequential depth. On the spectral side, K-Means clustering splits bands into redundant vs. informative groups, handled by lightweight CNN refinement and global Mamba modeling, respectively. On the spatial side, a layered overlapping spiral scan preserves 2-D continuity via shared quadrant encoding and an independent center encoder that stitches boundaries without corrupting the target pixel. An adaptive fusion gate further learns pixel-wise mixing weights to combine spectral and spatial cues. Experiments on three benchmarks demonstrate consistent improvements over recent Transformer- and Mamba-based baselines, especially in low-label regimes.**

***Index Terms*—Hyperspectral image classification, state space models (SSMs), Mamba, deep learning.**

I. INTRODUCTION

HYPERSPECTRAL imaging (HSI) is a key modality in Earth observation, capturing hundreds of narrow, contiguous spectral bands per scene [1]. Such dense sampling yields detailed spectral fingerprints, enabling reliable discrimination of materials that are indistinguishable in broadband multispectral imagery. Accordingly, HSI supports applications including precision agriculture [2], mineral mapping, and environmental monitoring [3]. However, this richness comes with high dimensionality and severe redundancy, which complicates learning under limited labels. The Hughes phenomenon [4]—accuracy degrading as dimensionality increases with limited training samples—together with inter-band redundancy and spatial heterogeneity, poses a central challenge for robust HSI feature extraction.

Early deep-learning approaches predominantly relied on CNNs, progressing from 2D backbones to multiscale designs [5], [6] and large-kernel variants [7] to expand the receptive field. To jointly model spectral–spatial cues, 3D CNNs (e.g., HybridSN [8]) were introduced and achieved strong performance, but remain biased toward local interactions. Convolution's locality bias limits long-range dependency modeling, which is critical for large contexts and class-ambiguous materials in remote-sensing scenes. Transformers with self-attention [9] were therefore adopted for global interactions; representative models (e.g., SpectralFormer [10] and SSFTT [11]) adapt ViT-style tokenization [12] to HSI. However, Transformer-based methods incur quadratic complexity in token length ($O(N^2)$) , resulting in prohibitive memory and computation, particularly for high-resolution HSIs with many bands.

Recently, state space models (SSMs), particularly Mamba [13], have emerged as an alternative that combines long-range modeling with linear-time complexity ($O(N)$). This efficiency has spurred Mamba-based vision models [14], [15] and subsequent HSI classifiers. Early HSI variants (e.g., Spectral-Spatial Mamba [16] and MambaHSI [17]) flatten HSI cubes into sequences for linear-time modeling. To better handle volumetric structure, 3DSS-Mamba [18] introduces 3D selective scanning, while S2Mamba [19] uses a dual-branch SSM with a mixture gate. To address spectral redundancy, IGroupSS-Mamba [20] uses interval grouping; SpiralMamba [21] adopts spiral scanning to mitigate spatial fragmentation.

Despite recent progress, current methods still struggle to decouple HSI's coupled spatial–spectral structure without sacrificing either long-range context or local fidelity. Most approaches oversimplify the spectral dimension in practice. They either model all bands as one long sequence—leading to state decay over hundreds of bands—or retain strict recurrence that limits parallelism, causing sequential latency. In the spatial domain, prevalent methods serialize a large patch using raster/spiral scanning. Such serialization creates a spatial–temporal gap: spatial neighbors may be far apart in the 1D order, weakening local correlations. As a result, the target (center) pixel can be dominated by background

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tokens; we refer to this as state pollution. Patch partitioning shortens paths but can place the target pixel near sub-patch boundaries, disrupting topology and introducing edge artifacts. Finally, static summation/concatenation for spectral–spatial fusion ignores class- and context-dependent cue dominance, limiting adaptability.

To address these challenges, we propose DTCMamba, a Dual-Branch Token-Chunking Mamba network for HSI classification. DTCMamba follows a divide-and-conquer design that decouples spectral redundancy and spatial topology while preserving long-range dependencies. First, we propose a history-enhanced parallel token-chunking SSM block (TCSSD) to mitigate state decay in long spectra and enable chunk-wise parallelism. Second, a cluster-based dual-branch spectral strategy separates redundant vs. informative bands and assigns them to lightweight CNN refinement vs. Mamba-based global modeling. Third, layered overlapping spiral scanning preserves 2D neighborhood structure and suppresses boundary artifacts via center-overlapping encoding. Finally, an adaptive fusion gate assigns pixel-wise weights to spectral vs. spatial features to handle class-dependent heterogeneity.

The main contributions of this article are summarized as follows:

1) We propose History-Enhanced Parallel TCSSD, an SSM block that injects gated second-order history and supports chunk-wise parallel processing to mitigate state decay and reduce sequential depth.
2) We design a cluster-based dual-branch spectral module that separates redundant vs. informative bands (via K-Means) and models them with lightweight CNN refinement vs. Mamba-based global modeling, improving efficiency without sacrificing accuracy.
3) We introduce layered overlapping spiral scanning with a shared quadrant encoder and an independent center encoder, preserving topology while alleviating boundary artifacts around the target pixel.
4) We develop a pixel-wise adaptive fusion gate to weight spectral vs. spatial features, and demonstrate consistent gains on Indian Pines, Houston 2013, and WHU-Hi-LongKou, especially under limited training samples.

II. RELATED WORKS

A. CNN-based HSI Classification

Hyperspectral image (HSI) classification aims to assign a specific land-cover category to each pixel by analyzing its continuous spectral signature and spatial context. Over the past decades, methods for HSI classification have evolved from traditional machine learning approaches to advanced deep learning frameworks.

Early shallow classifiers (e.g., SVMs [22]) struggled with complex non-linear spectral mappings. CNNs subsequently dominated, evolving from simple 1-D/2-D extractors to unified 3-D spatial-spectral frameworks (e.g., SSRN [23]) and lightweight variants [24]. To enhance feature discriminability, later works integrated spatial attention mechanisms [25], multi-view fusion strategies [26], and graph-based weighting [27] into convolutional backbones. Hybrid architectures like HybridSN [8] further optimize the trade-off between 3-D and 2-D operations. However, the intrinsic local receptive field of convolutions limits the capture of long-range global dependencies essential for complex HSI scenes.

B. Transformer-based HSI Classification

To alleviate CNN locality, Transformers with self-attention [9], [12] were introduced to model global token interactions. Hierarchical designs such as Swin [28], [29] motivate window-based HSI models that trade off local aggregation and global context. SpectralFormer [10] and SSFTT [11] adapt ViT to HSI by tokenizing spectral bands and/or spatial patches. Later works inject locality priors into Transformers to strengthen fine-grained feature extraction. For instance, HiT [30] embeds 3-D convolutions into Transformers to capture fine spectral–spatial cues. DCTN [31] and SS-TMNet [32] use dual streams to combine CNN local features with Transformer global dependencies. Lightweight interactive Transformers [33] reduce overhead by restricting attention patterns. In addition, recent Scale-interaction Transformer (SiT [34]) further introduces multi-scale interaction into Transformer-based HSI classification, enhancing the representation of spatial-spectral structures through scale-aware feature modeling. To mitigate label scarcity, self-supervised learning (e.g., mask-enhanced contrastive learning [35]) and cross-domain adaptation [36] leverages unlabeled HSI for representation pretraining. Despite stronger global modeling, Transformers incur quadratic complexity ($O(N^2)$) in token length, leading to high memory and compute costs for high-resolution HSIs with many bands.

C. Mamba-based HSI Classification methods

Recently, SSMs, particularly Mamba [13], have emerged as an alternative to Transformers for efficient long-range modeling. Mamba uses selective scan to model long-range dependencies with linear-time complexity ($O(N)$), which is attractive for HSI sequences. In general vision, VMamba [14] and Vision Mamba [15] show competitive performance with ViTs while improving efficiency. Mamba variants for video understanding [37] and spectral compressive imaging [38] further suggest applicability to 3-D data, which is structurally related to HSI cubes. Recent work increasingly explores Mamba for HSI classification. Spectral-Spatial Mamba [16] and MambaHSI [17] flatten HSI cubes into sequences for linear-time modeling, often achieving accuracy comparable to Transformers with lower overhead. 3DSS-Mamba [18] introduces 3-D selective scanning to extract spectral–spatial features directly from cubes, and S2Mamba [19] uses a dual-branch architecture with a spatial–spectral mixture gate for interactive fusion. Several specialized variants further adapt Mamba to HSI characteristics: IGroupSS-Mamba [20] mitigates spectral redundancy via interval grouping and group-wise modeling, SpiralMamba [21] uses spiral scanning to reduce spatial fragmentation and preserve local continuity, and LSFMamba [39] employs local-enhanced spiral fusion for robust land cover classification. However, challenges remain in decoupling spectral redundancy without degrading long-range

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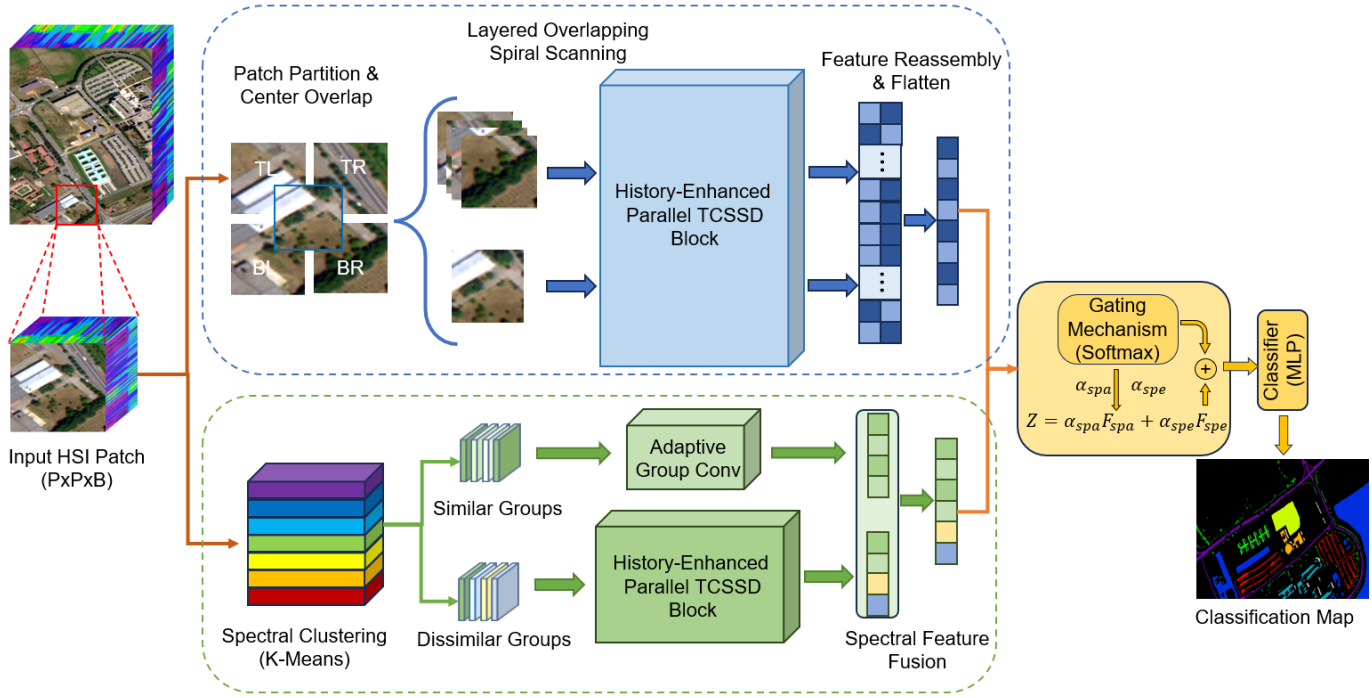


Fig. 1: The overall architecture of the proposed Dual-Branch Token-Chunking Mamba (DTCMamba). The model processes the HSI patch through two parallel streams: the Cluster-Based Spectral Branch (bottom) and the Layered Overlapping Spiral Spatial Branch (top). Both branches utilize the History-Enhanced Parallel TCSSD Block as the core feature extractor.

dependency modeling, and in mitigating boundary artifacts during spatial serialization; DTCMamba addresses these via history-augmented states and layered overlapping scanning.

Existing Mamba-based HSI methods typically emphasize either spectral efficiency (e.g., grouping) or spatial serialization (e.g., spiral scan), but seldom address state decay under ultra-long spectra, parallelizable sequence modeling, and boundary-robust target encoding in a unified design. Our DTCMamba explicitly targets these three aspects via history-augmented token–chunking SSMs, cluster-based spectral decoupling, and layered overlapping spatial scanning.

III. PROPOSED METHOD

A. Preliminaries

1) State Space Models (SSMs) serve as a foundational framework for modeling continuous systems, mapping a 1-D input sequence $x(t) \in \mathbb{R}^L$ to an output sequence $y(t) \in \mathbb{R}^L$ through a latent state $h(t) \in \mathbb{R}^N$. The continuous-time evolution is governed by the following ordinary differential equations (ODEs):

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) + Dx(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$represents the state transition matrix, and $\mathbf{B} \in \mathbb{R}^{N\times L}, \mathbf{C} \in \mathbb{R}^{L\times N}$, are projection parameters, $\mathbf{D}$ is usually served as a residual connected operation. To apply this continuous system to discrete HSI data, the Zero-Order Hold (ZOH) method is typically employed to discretize the parameters using a timescale parameter Δ.

$$\bar{\mathbf{A}} = \exp(\mathbf{\Delta A}), \quad \bar{\mathbf{B}} = (\mathbf{\Delta A})^{-1}(\exp(\mathbf{\Delta A}) - \mathbf{I}) \cdot \mathbf{\Delta B} \tag{2}$$

where $\overline{A}$ and $\overline{B}$ represent the discretized forms of the parameters $\mathbf{A}$ and $\mathbf{B}$. The discretized parameters $\overline{A}$ and $\overline{B}$ allow the model to be computed recurrently:

$$\begin{aligned} h_t &= \overline{A}h_{t-1} + \overline{B}x_t \\ y_t &= Ch_t \end{aligned} \tag{3}$$

2) Standard SSMs are linear time-invariant (LTI), which limits their ability to select relevant information from the complex spectral signatures of HSI. Mamba addresses this by introducing a selective scan mechanism where parameters $\mathbf{B}$, $\mathbf{C}$, and Δ are input-dependent. However, for ultra-long hyperspectral sequences, the standard selective scan may still suffer from information decay over time.

In HSI classification, such information decay may weaken discriminative long-range spectral cues. Since an HSI pixel usually contains hundreds of contiguous bands, material-specific absorption or reflection responses may appear at distant wavelength positions. In the discrete SSM recurrence, the contribution of an earlier band to a later hidden state is propagated through repeated state transitions, e.g., approximately related to $\bar{A}^{t-i}\bar{B}x_i$, and may gradually attenuate as the spectral distance increases. Consequently, the final representation can be biased toward nearby or recently scanned bands, reducing the separability of land-cover classes with similar local spectral patterns. This motivates our history-enhanced TCSSD block, which introduces a gated second-order historical state to alleviate long-range spectral forgetting.

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B. DTCMamba Overview

As illustrated in Fig. 1, the proposed DTCMamba is designed to efficiently extract and integrate discriminative features from the complex spatial-spectral domain. The overall workflow comprises four stages: (1) input embedding, (2) dual-branch feature extraction, (3) adaptive fusion, and (4) classification. The dual-branch feature extraction comprises three key components: (i) History-Enhanced Parallel TCSSD Block, (ii) Cluster-Based Spectral Branch (Bottom Stream), and (iii) Layered Overlapping Spiral Spatial Branch (Top Stream).

1) Input Embedding: Given a hyperspectral image $I \in \mathbb{R}^{H\times W\times B_{\text{org}}}$, we first extract a 3-D patch $X \in \mathbb{R}^{P\times P\times B_{\text{org}}}$centered at the target pixel. A preliminary 1×1 convolution layer is applied to reduce the spectral dimension to B, generating the initial feature embedding.

2) History-Enhanced Parallel TCSSD Block: To address spectral–spatial heterogeneity, we split the embedding into two parallel branches; both branches are instantiated with the same History-Enhanced Parallel TCSSD block to model sequence dependencies.

3) Cluster-Based Spectral Branch (Bottom Stream) : The spectral branch targets long-range spectral signatures via cluster-guided decoupling. It employs K-Means [40] to partition bands into high-redundancy (similar) groups, refined by lightweight CNN denoising, and high-information (dissimilar) groups, modeled by the Parallel TCSSD block to capture global dependencies.

4) Layered Overlapping Spiral Spatial Branch (Top Stream): Complementing Spectral Branch, the Layered Overlapping Spiral Spatial Branch focuses on 2-D spatial topology through a specific scanning strategy that partitions the patch into four quadrants and a center-overlapping block; these sub-blocks are processed using parallel spiral scanning and the Parallel TCSSD Block encoding, thereby preserving spatial neighborhood structures and effectively alleviating boundary artifacts.

5) Dual-Branch Adaptive Fusion: The outputs from the spatial and spectral branches are heterogeneous. As shown in the right part of Fig. 1, they are fed into a Dual-Branch Adaptive Fusion gate. This module calculates a pixel-wise weight vector via a multi-layer perceptron (MLP) and Softmax operation, allowing the model to adaptively emphasize either spatial texture or spectral material properties based on the specific land-cover class.

6) Classification: Finally, the fused feature map is flattened and passed through a classifier head, comprising Global Average Pooling and Fully Connected layers, to predict the probability distribution over the C land-cover categories.

C. History-Enhanced Parallel TCSSD

Standard Mamba architectures rely on a continuous-time system that maps a 1-D input $x(t) \in R$ to an output $y(t) \in R$ through a latent state $h(t) \in R^N$. To adapt this to discrete image sequences, we introduce the History-Enhanced Parallel TCSSD module. The internal microscopic mechanism of this unit is visually detailed in Fig. 2.

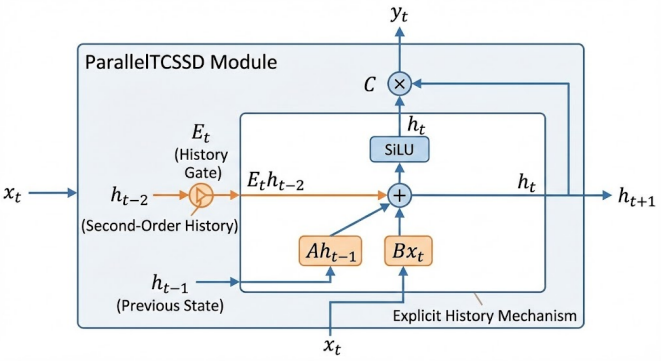


Fig. 2: The detailed structure of the proposed History-Enhanced Parallel TCSSD Module. Unlike standard SSMs that only rely on the previous state h_{t-1}, we introduce an explicit Second-Order History mechanism. A learnable History Gate (E_t) controls the injection of the ante-penultimate state h_{t-2}, ensuring robust long-term dependency modeling.

First, consistent with the standard SSM, we transform the continuous parameters (A, B) into discrete parameters $(\overline{A}, \overline{B})$ using the Zero-Order Hold (ZOH) method with a timescale parameter Δ. This process allows the model to capture sequence dependencies at a specific resolution, as shown in Equation (Eq. 2). A is the state transition matrix and B is the projection matrix. In our design, these discrete parameters govern the intra-chunk local dependencies.

Standard SSMs strictly follow a first-order Markovian recurrence where h_t depends solely on h_{t-1}, which often leads to information decay (the forgetting issue) when processing long spectral sequences. To address this, we fundamentally redesign the state equation.

As illustrated in the "Explicit History Mechanism" block of Fig. 2, we introduce a Second-Order History path. Specifically, we define a history term h_{t-1}^{hist} which acts as a memory anchor. In our specific implementation shown in the figure, this term fetches the ante-penultimate state, $h_{t-1}^{hist} \leftarrow h_{t-2}$ to create a "short-circuit" for gradient flow.

Crucially, to reduce the excessive influence of historical information on the current context, we introduce a learnable History Gate, denoted as E_t. This gate dynamically regulates the injection of the historical state. The modified state update rule is formulated as:

$$\begin{aligned} h_t &= \sigma\left(\overline{\mathbf{A}} h_{t-1} + \overline{\mathbf{B}} x_t + \mathbf{E}_t \odot h_{t-1}^{hist}\right) \\ &= \text{SiLU}\left(\overline{\mathbf{A}} h_{t-1} + \overline{\mathbf{B}} x_t + \mathbf{E}_t \odot h_{t-2}\right) \end{aligned} \tag{4}$$

Here, σ represents the activation function, implemented as SiLU in Fig. 2. The term $E_t \odot h_{t-2}$ explicitly represents the gated second-order history injection, where E_t is a data-dependent parameter learned alongside A and B. This mechanism helps alleviate the vanishing gradient problem in long sequences by providing an auxiliary temporal bridge.

The gated second-order term provides an auxiliary propagation path for historical spectral states. In a standard first-order SSM recurrence, distant spectral information is repeatedly transformed through consecutive state transitions and may gradually decay. By directly injecting h_{t-2} into the current

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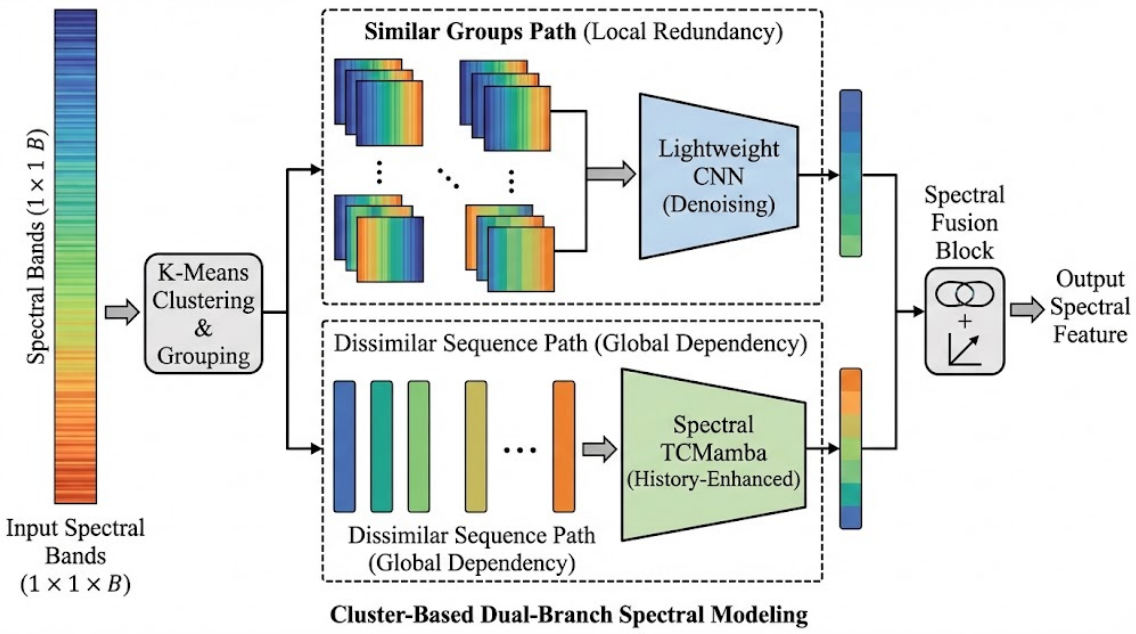


Fig. 3: The architecture of the proposed Cluster-Based Dual-Branch Spectral Modeling. The module dynamically decouples the input spectral bands into Similar and Dissimilar paths via K-Means. The Similar Path employs CNNs for local denoising, while the Dissimilar Path utilizes Spectral TCMamba for global dependency modeling.

update, the proposed mechanism shortens the propagation path of historical information and improves gradient flow for long spectral sequences. Different from a simple residual connection or an ungated memory buffer, the learnable gate E_t adaptively controls the injected historical information, allowing the model to preserve useful long-range spectral cues while suppressing redundant or noisy past states.

Finally, the updated state h_t is used to generate the current output. As depicted in the outer loop of Fig. 2, the module employs a multiplicative gating branch to modulate the feature flow. The final output y_t of the TCSSD block is computed as:

$$y_t = (\mathbf{C}h_t) \odot \sigma(\mathbf{D}x_t) \tag{5}$$

where C is the output projection matrix and D represents the direct residual connection.

Furthermore, unlike the full-length sequential recurrence in vanilla Mamba, our Parallel TCSSD partitions a sequence of length L into approximately $\lceil L/C \rceil$ chunks, where C denotes the number of tokens in each chunk. The resulting chunks are processed in parallel, reducing the longest within-chunk recurrent path from L to C. This chunk-wise execution improves GPU utilization, while the History Equation (Eq. 4) helps preserve historical information affected by sequence partitioning.

D. Cluster-Based Dual-Branch Spectral Modeling

Directly modeling all B bands as a single sequence often leads to computational inefficiency due to the high redundancy between adjacent bands. To address this, we propose a Cluster-Based Dual-Branch strategy. The overall architecture of this module is illustrated in Fig. 3. The module receives the raw spectral sequence ($1 \times 1 \times B$) as input and processes it through the following key stages:

1) K-Means Spectral Grouping: As shown in the left part of Fig. 3, we utilize the K-Means algorithm to dynamically partition all spectral bands into multiple clusters based on spectral similarity. Let $S = b_1, b_2, \ldots, b_l$ be the set of spectral bands. The clustering generates sets of indices C_k for $k = 1 \ldots K$. This step naturally decouples the bands into two distinct categories: Similar and Dissimilar .

2) Similar Branch : Bands within the same cluster (C_k) are highly correlated and often contain redundant noise. Modeling these with SSM is computationally wasteful. In the upper branch of Fig. 3, bands that are highly correlated are assigned to the same groups (visualized as clusters with similar colors). These groups contain substantial local redundant information. Consequently, they are fed into a lightweight CNN module for efficient local denoising and feature extraction, avoiding unnecessary computational overhead on redundant data.

3) Dissimilar Branch : In the lower branch, to capture global spectral dependencies, representative bands from different clusters (visualized as discrete bands with distinct colors) are selected to form a new "Dissimilar Sequence." This sequence exhibits high variance and represents the global spectral dependencies of the material. It is processed by the Spectral TCMamba module equipped with the proposed history-enhanced mechanism to model long-range interactions.

Finally, the local features from the CNN branch and the global features from the TCMamba branch are merged in the Spectral Fusion module to generate the final comprehensive spectral feature representation.

E. Layered Overlapping Spiral Scanning

To effectively mitigate the spatial-temporal gap and state pollution inherent in standard raster serialization, we propose

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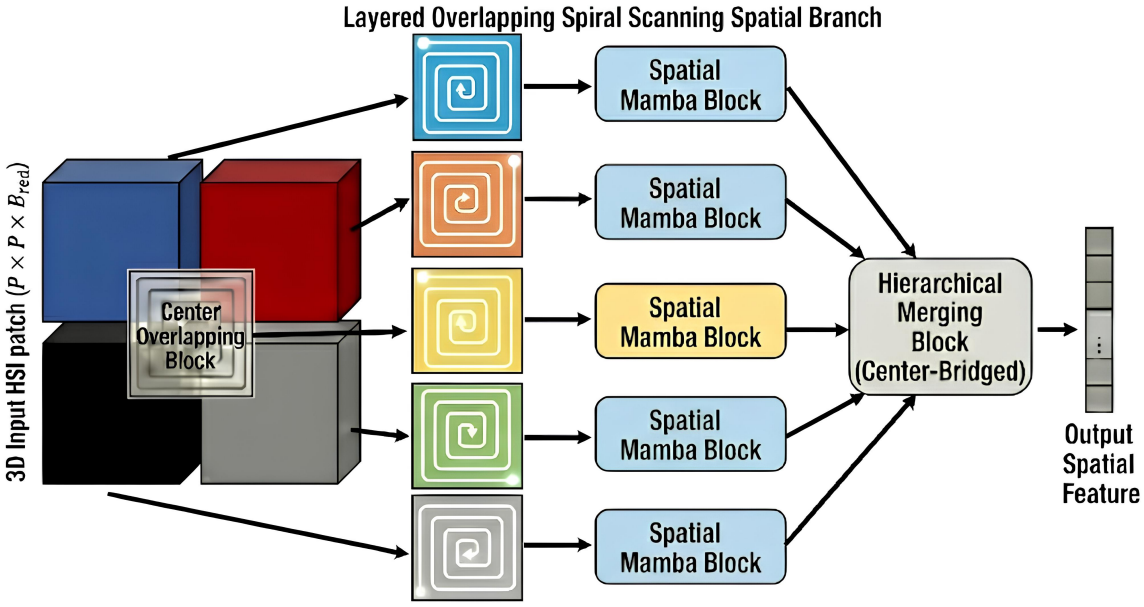


Fig. 4: Layered overlapping spiral scanning. Four quadrant sequences are encoded by a parameter-shared Spatial Mamba (Θ), whereas the center-overlapping sequence uses an independent encoder (Θ_u). Hierarchical merging uses the center feature to bridge quadrant boundaries and refine the target-pixel representation.

the Layered Overlapping Spiral Scanning module. This architecture, as visually detailed in Fig. 4 orchestrates a transition from 2-D images to 1-D sequences through a physically meaningful "divide-and-conquer" protocol. Let the input patch be denoted as $\mathbf{X} \in \mathbb{R}^{P\times P\times B}$. Instead of flattening $\mathbf{X}$ directly, we first partition it into five strategic sub-regions: four non-overlapping corner quadrants $\{\mathbf{X}_{TL}, \mathbf{X}_{TR}, \mathbf{X}_{BL}, \mathbf{X}_{BR}\}$ representing the background context, and one Center Overlapping Block $\mathbf{X}_C$ that covers the crucial intersection area. Mathematically, if the patch size is $P \times P$, the quadrants have a size of $\frac{P}{2} \times \frac{P}{2}$, while the center block is defined as the block enclosed by the center pixels of the four angular quadrants, creating a spatial bridge across the partition boundaries. Subsequently, a continuous spiral scanning function $\mathcal{S}(\cdot)$ maps each 2-D sub-block into a 1-D sequence, preserving local neighborhood adjacency more effectively than raster scanning.

The core innovation lies in the Dual-Mechanism Encoding strategy tailored for these sequences. As illustrated in the parallel processing stage of Fig. 4 , the four quadrant sequences are grouped and fed into a Common Spatial Mamba (visualized as blue modules). From a coding perspective, these four sub-patches are stacked along the batch dimension to form a tensor of shape $(4B, \frac{L}{4}, B)$. They share the same set of learnable parameters $\mathbf{\Theta}$, forcing the model to learn position-invariant statistical features of the background texture (Context Compactness). This weight-sharing design significantly reduces parameter redundancy. In contrast, the sequence derived from the Center Overlapping Block is directed to a dedicated Independent Spatial Mamba (visualized as a distinct orange module with parameters $\mathbf{\Theta}_u$). This branch operates independently to perform receptive field re-centering. Since the center block explicitly contains the target pixel and its immediate neighbors, the independent parameters $\mathbf{\Theta}_u$ allow the model to specialize in fine-grained target refinement without being diluted by the broad background context. Thus, the five Spatial Mamba blocks illustrated in Fig. 4 correspond to the five generated spatial sequences rather than five fully independent parameter sets. Specifically, the four quadrant branches share the same Common Spatial Mamba parameters Θ to learn consistent and position-invariant contextual patterns, while the center branch adopts independent parameters Θ_u to focus on target-pixel refinement.

Finally, the encoded features are reconstructed via a Hierarchical Merging process. Let $\mathbf{F}_{\text{quad}} \in \mathbb{R}^{P\times P\times B}$ denote the spatial map reconstructed from the shared branch, which often suffers from blocking artifacts at the quadrant boundaries (the "cross" lines). We overlay the features $\mathbf{F}_{center}$ generated by the independent branch onto the central region of $\mathbf{F}_{quad}$. This operation effectively acts as a "spatial stitcher," smoothing out the discontinuities caused by partitioning and ensuring that the classification target—located at the geometric center—is represented by a seamless, high-integrity feature vector. This hierarchical design alleviates the trade-off between global context modeling and local feature precision, yielding a robust spatial representation.

$$Y_{spa} = \mathcal{H}\left(\bigcup_{k\in\Omega} \text{SSM}(\mathcal{S}(\mathbf{X}_k); \mathbf{\Theta}),\ \text{SSM}(\mathcal{S}(\mathbf{X}_C); \mathbf{\Theta}_u)\right) \tag{6}$$

where $\Omega = TL, TR, BL, BR$ represents the set of four corner quadrants, and $\mathcal{S}(\cdot)$ denotes the spiral scanning function. $\mathbf{\Theta}$ refers to the weight-shared parameters of the Common Spatial Mamba for background consistency, while $\mathbf{\Theta}_u$ represents the parameters of the Independent Spatial Mamba (orange module) for center target refinement. $\mathcal{H}$ signifies the hierarchical spatial stitching operation.

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F. Dual-Branch Adaptive Fusion

Spectral and spatial features contribute differently depending on the land-cover class (e.g., shape is crucial for "Roads," while spectral signature is crucial for "Corn"). To avoid fixed-weight summation, we propose a Dual-Branch Adaptive Fusion gate.

Let F_{spe} and F_{spa} be the outputs of the spectral and spatial branches, respectively. We compute a dynamic attention weight vector $\alpha \in \mathbb{R}^2$:

$$\alpha = \mathrm{Softmax}(\mathrm{MLP}([F_{\mathrm{spe}}, F_{\mathrm{spa}}])) \tag{7}$$

The final fused feature Z is obtained by:

$$Z = \alpha_{spa} \cdot F_{spa} + \alpha_{spe} \cdot F_{spe} + X_{res} \tag{8}$$

This mechanism encourages competition between spatial and spectral features, allowing the model to adaptively focus on the most discriminative domain for each pixel.

G. Computational Complexity Analysis

Let the input patch size be $P \times P$, the number of spatial tokens be $N_s = P^2$, the number of spectral bands be B, and the hidden feature dimension be d. For attention-based models, the computational cost grows quadratically with the sequence length, e.g., $O(N_s^2 d)$ for spatial tokens or $O(B^2 d)$ for spectral tokens. In contrast, the SSM-based spatial and spectral modeling in DTCMamba scales linearly with the processed sequence length, i.e., approximately $O(N_s d)$ and $O(Bd)$, respectively.

Nevertheless, the actual implementation also contains several auxiliary costs, including K-Means spectral grouping, lightweight convolutional refinement for similar-band groups, center-overlapping spatial encoding, and parallel chunk-state maintenance. Specifically, K-Means grouping introduces a cost of approximately $O(IKBN_s)$, where I is the number of K-Means iterations and K is the number of clusters. The lightweight CNN branch and chunk-state maintenance introduce additional overheads, denoted as $O(\mathcal{C}_{\mathrm{conv}})$ and $O(\mathcal{C}_{\mathrm{chunk}})$, respectively. Therefore, the overall complexity can be written as

$$\begin{aligned}\mathcal{O}_{\mathrm{DTCMamba}} = &O(IKBN_s) + O(Bd) + O(N_s d) \\ &+ O(\mathcal{C}_{\mathrm{conv}}) + O(\mathcal{C}_{\mathrm{chunk}}).\end{aligned} \tag{9}$$

Although DTCMamba includes these auxiliary operations, its dominant spatial and spectral sequence modeling terms remain linear rather than quadratic, which preserves the efficiency advantage over Transformer-style attention under comparable token lengths.

IV. DATASETS AND EVALUATION METRICS

In this section, we will present the datasets and evaluation criteria employed in the experiments.

A. Datasets

We conduct evaluations of our DTCMamba on three publicly available datasets, focusing on HSI classification: Indian Pines, Houston 2013, and WHU-Hi-LongKou datasets.

1) Indian Pines: The IP dataset [41] consists of 145 × 145 pixels at a ground sampling distance of 20 m and 220 spectral bands spanning the wavelength range of 400–2500 nm, in which 200 spectral bands are preserved after removing 20 noisy and water absorption bands. The dataset is annotated using 16 land cover categories, including crops, trees, and other vegetation. The details are exhibited in Table I.

TABLE I: LAND-COVER CATEGORIES OF THE IP DATASET AND ITS CORRESPONDING TRAINING AND TESTING SAMPLES

No.	Category	Training	Testing	Total
1	Alfalfa	5	41	46
2	Corn-notill	143	1285	1428
3	Corn-mintill	83	747	830
4	Corn	24	213	237
5	Grass-pasture	48	435	483
6	Grass-trees	73	657	730
7	Grass-pasture-mowed	3	25	28
8	Hay-windrowed	48	430	478
9	Oats	2	18	20
10	Soybean-notill	97	875	972
11	Soybean-mintill	245	2210	2455
12	Soybean-clean	59	534	593
13	Wheat	20	185	205
14	Woods	126	1139	1265
15	Buildings-Grass-Trees-Drives	39	347	386
16	Stone-Steel-Towers	9	84	93
	Total	1024	9225	10249

2) Houston 2013: The Houston 2013 dataset [42] consists of 349 × 1905 pixels at a ground sampling distance of 2.5 m and 144 spectral bands spanning the wavelength range of 380–1050 nm. The dataset is annotated using 15 land cover categories, including healthy grass, stressed grass, synthetic grass, trees, soil, water, residential, and others. The details are exhibited in Table II.

TABLE II: LAND-COVER CATEGORIES OF THE HOUSTON 2013 DATASET AND ITS CORRESPONDING TRAINING AND TESTING SAMPLES

No.	Category	Training	Testing	Total
1	Healthy Grass	13	1238	1251
2	Stressed Grass	13	1241	1254
3	Synthetic Grass	7	690	697
4	Tree	12	1232	1244
5	Soil	12	1230	1242
6	Water	3	322	325
7	Residential	13	1255	1268
8	Commercial	12	1232	1244
9	Road	13	1239	1252
10	Highway	12	1215	1227
11	Railway	12	1223	1235
12	Parking Lot1	12	1221	1233
13	Parking Lot2	5	464	469
14	Tennis Court	4	424	428
15	Running Track	7	653	660
	Total	150	14879	15029

3) WHU-Hi-LongKou: The dataset [43] consists of 550 × 400 pixels along with 270 spectral bands spanning the wavelength range of 0.4–1.0 μm. The dataset is annotated using nine land cover categories, including corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, rice, water, roads and

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houses, and mixed weed. The details are exhibited in Table III.

TABLE III: LAND-COVER CATEGORIES OF THE WHU-HI-LONGKOU DATASET AND ITS CORRESPONDING TRAINING AND TESTING SAMPLES

No.	Category	Training	Testing	Total
1	Corn	17	34494	34511
2	Cotton	4	8370	8374
3	Sesame	1	3030	3031
4	Broad-Leaf Soybean	32	63180	63212
5	Narrow-Leaf Soybean	2	4149	4151
6	Rice	6	11848	11854
7	Water	33	67023	67056
8	Roads and Houses	4	7120	7124
9	Mixed Weed	3	5226	5229
	Total	102	204440	204542

B. Evaluation Metrics

In this article, we utilized three commonly used evaluation metrics: OA, average accuracy (AA), and kappa coefficient (κ) to evaluate the classification performance of our DTC-Mamba. OA is the overall accuracy of all categories, which aims to evaluate the model's overall performance. AA is the average accuracy of each category, which reflects the average performance of each category and the degree of category imbalance. Kappa coefficient is a measure of observed and expected consistency.

V. EXPERIMENT

A. Implementation Details

In this study, all experiments were conducted within the PyTorch framework [44], using 8 NVIDIA A40 GPUs, each equipped with 48 GB of memory. For the initialization of our DTCMamba, its parameters are randomly initialized by a zero-mean normal distribution with a standard deviation of 0.01. Adam [45] is adopted as the optimizer for DTCMamba, where the exponential learning rate strategy is applied with an initial value of 0.001. The model is trained with 100 epochs, where the batch size is set to 64. The patch size of inputs is set to 15, 15, and 15 for IP, Houston 2013, and WHU-HiLongKou, respectively. The PCA dimension for reduction is set to 30. The spectral grouping size was set to 4. In addition, the optimal parallel chunk size was set to 8. The effects of image patch size, chunk size and spectral group number will be analyzed in detail in Section V-D. All the experiments are performed under identical experimental conditions, with the reported results averaged over five consecutive trials.

B. Comparison With State of the Arts

To verify the superiority of the proposed method, we compare DTCMamba with several state-of-the-art architectures, covering both advanced Transformer-based models (HiT [30], SSFTT [11], SS-TMNet [32], DCTN [31], and SiT [34]) and recent Mamba-based models (SS-Mamba [16], MambaHSI [17], 3DSS-Mamba [18], IGroupSS-Mamba [20], and S2Mamba [19]). All methods were evaluated using the optimal settings reported in their respective studies.

1) Indian Pines: Table IV reports the classification results on the IP dataset with a 10% sampling ratio. Among the Transformer-based methods, SiT achieves the most competitive performance, with an OA of 96.84%, which is significantly higher than HiT, SSFTT, SS-TMNet, and DCTN. This indicates that scale-interaction modeling can effectively enhance spatial-spectral feature representation. Compared with Transformer-based methods, Mamba-based methods generally obtain stronger performance under the same training setting. In particular, IGroupSS-Mamba and S2Mamba both achieve an OA of 98.67%, showing the effectiveness of spectral grouping and spatial-spectral state modeling for HSI classification. Nevertheless, the proposed DTCMamba achieves the highest OA of 98.97%, AA of 98.84%, and Kappa of 98.82%. Compared with the strongest competing Mamba-based methods, DTCMamba improves the OA by 0.30%. The improvement can be attributed to the cluster-based spectral decoupling strategy and the history-enhanced TCSSD block, which jointly reduce spectral redundancy and strengthen long-range spectral dependency modeling. In addition, the layered overlapping spiral scanning further preserves spatial topology, leading to more discriminative spectral-spatial representations for similar crop categories.

To further illustrate the classification performance, Fig. 5 presents the visualization maps of various methods. It can be observed that HiT, SSFTT and DCTN generally suffer from significant noise and boundary confusion. In contrast, the proposed DTCMamba demonstrates the most consistent results with the Ground Truth, achieving clearer category boundaries and smoother homogeneous regions compared to other Mamba-based competitors.

2) Houston 2013: Experiments on the Houston 2013 dataset with a 1% sampling ratio further verify the robustness of the proposed method in complex urban scenarios, as shown in Table V. Among the Transformer-based methods, SiT obtains the highest OA of 89.82%, outperforming the other Transformer baselines. This suggests that multi-scale interaction is beneficial for modeling urban spatial structures. Among Mamba-based methods, S2Mamba achieves the best competing OA of 90.47%, followed by IGroupSS-Mamba with an OA of 90.08%. However, DTCMamba achieves a remarkable OA of 93.23%, surpassing S2Mamba by 2.76%. DTCMamba also obtains the highest AA of 93.31% and Kappa of 92.68%, indicating better overall classification consistency. This performance gain mainly benefits from the dual-branch design, where the spectral branch adaptively separates redundant and informative bands, while the spatial branch preserves local topology through layered overlapping spiral scanning. These designs are particularly effective for urban scenes with complex object boundaries and high intra-class spectral variability.

To intuitively evaluate the qualitative performance, Fig. 6 displays the classification maps on the Houston 2013 dataset. While 3DSS-Mamba struggles with spatial continuity, the proposed DTCMamba achieves the smoothest visualization with well-defined boundaries, effectively reducing the misclassification of shadow and building classes compared to other competitors.

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TABLE IV: Comparison with State-of-the-Art methods on the Indian Pines dataset (10% training samples). The best results are in bold.

Class	Transformers					Mamba					Ours
	HiT	SSFTT	SS-TMNet	DCTN	SiT	MambaHSI	3DSS-Mamba	SS-Mamba	IGroupSS-Mamba	S2Mamba	DTCMamba
1	69.05 ± 16.15	97.56 ± 1.72	75.71 ± 12.19	90.95 ± 9.88	84.88 ± 12.72	**100.00 ± 0.00**	79.02 ± 19.71	67.32 ± 17.13	93.66 ± 1.34	94.63 ± 6.54	**100.00 ± 0.00**
2	84.70 ± 4.84	92.51 ± 10.58	89.67 ± 2.96	93.73 ± 2.34	91.86 ± 0.39	95.24 ± 0.27	93.06 ± 1.18	92.03 ± 3.37	**96.16 ± 1.00**	94.72 ± 0.59	95.67 ± 0.23
3	59.79 ± 7.56	89.69 ± 12.29	69.32 ± 3.65	73.41 ± 0.68	97.32 ± 1.10	99.14 ± 0.31	97.75 ± 1.00	97.22 ± 2.15	99.73 ± 0.25	**99.92 ± 0.18**	**99.92 ± 0.12**
4	59.91 ± 13.50	85.07 ± 21.60	74.39 ± 10.81	81.40 ± 3.14	97.65 ± 2.48	**99.62 ± 0.21**	88.08 ± 7.57	89.39 ± 5.91	99.53 ± 0.57	99.72 ± 0.42	99.34 ± 0.54
5	69.20 ± 1.99	77.29 ± 40.56	73.97 ± 1.23	76.87 ± 2.67	95.03 ± 2.34	98.25 ± 3.53	93.15 ± 4.58	95.49 ± 0.90	98.80 ± 0.72	99.08 ± 0.28	**98.94 ± 0.70**
6	96.89 ± 1.80	92.66 ± 9.35	97.99 ± 1.14	99.18 ± 1.08	99.15 ± 0.61	99.82 ± 0.20	95.16 ± 5.32	99.18 ± 0.66	99.60 ± 0.17	**100.00 ± 0.00**	99.79 ± 0.14
7	66.92 ± 28.08	80.00 ± 44.72	56.15 ± 28.86	62.31 ± 16.63	88.00 ± 4.90	**100.00 ± 0.00**	44.00 ± 39.70	93.60 ± 10.43	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
8	89.19 ± 0.35	96.09 ± 8.61	89.51 ± 0.50	89.61 ± 0.64	**99.95 ± 0.10**	**100.00 ± 0.00**	97.40 ± 2.85	99.95 ± 0.10	99.95 ± 0.10	**100.00 ± 0.00**	**100.00 ± 0.00**
9	62.22 ± 37.35	45.56 ± 46.21	70.00 ± 18.67	45.55 ± 9.13	90.00 ± 10.69	80.00 ± 11.52	43.33 ± 23.04	57.78 ± 16.01	83.33 ± 20.41	80.00 ± 15.52	**94.44 ± 5.56**
10	78.42 ± 1.47	78.22 ± 26.83	82.10 ± 2.92	88.09 ± 1.65	95.93 ± 0.81	**99.43 ± 0.21**	94.33 ± 4.35	96.50 ± 2.11	98.67 ± 0.57	99.04 ± 0.38	99.38 ± 0.13
11	88.37 ± 1.70	98.65 ± 0.33	89.47 ± 2.45	92.84 ± 0.49	98.71 ± 0.47	99.18 ± 0.69	97.98 ± 0.46	99.00 ± 0.46	99.36 ± 0.38	99.38 ± 0.19	**99.67 ± 0.18**
12	73.56 ± 3.90	92.58 ± 6.29	81.84 ± 3.00	87.94 ± 1.72	91.84 ± 1.48	95.28 ± 3.62	92.25 ± 3.60	95.51 ± 1.90	96.10 ± 1.48	96.33 ± 1.50	**97.45 ± 0.63**
13	95.03 ± 3.05	89.62 ± 22.01	**99.89 ± 0.24**	99.78 ± 0.48	99.68 ± 0.48	98.92 ± 0.66	91.68 ± 5.95	98.59 ± 0.30	99.78 ± 0.30	99.35 ± 0.89	99.35 ± 0.45
14	94.91 ± 1.44	86.57 ± 29.94	96.45 ± 0.33	96.52 ± 0.42	99.84 ± 0.22	99.93 ± 0.11	98.51 ± 2.15	99.46 ± 0.65	99.98 ± 0.04	**100.00 ± 0.00**	**100.00 ± 0.00**
15	42.47 ± 5.60	90.43 ± 11.70	50.63 ± 2.52	53.45 ± 2.24	97.93 ± 1.45	**99.83 ± 0.16**	89.05 ± 8.49	93.78 ± 4.04	98.96 ± 1.13	99.94 ± 0.13	**99.83 ± 0.16**
16	80.00 ± 18.28	81.90 ± 9.16	90.48 ± 6.89	93.33 ± 4.50	92.14 ± 2.16	86.43 ± 2.99	80.71 ± 10.73	90.24 ± 3.30	92.62 ± 2.58	98.81 ± 1.68	**97.62 ± 1.19**
OA (%)	81.76 ± 2.01	90.62 ± 10.30	85.68 ± 1.01	88.87 ± 0.49	96.84 ± 0.28	98.44 ± 0.53	95.02 ± 1.57	96.65 ± 1.18	98.67 ± 0.19	98.67 ± 0.17	**98.97 ± 0.03**
AA (%)	75.66 ± 5.86	85.90 ± 11.73	80.47 ± 3.08	82.81 ± 1.06	94.99 ± 0.70	96.94 ± 0.49	85.97 ± 6.05	91.56 ± 2.18	97.27 ± 1.38	97.56 ± 0.76	**98.84 ± 0.25**
$\kappa \times 100$	79.33 ± 2.30	89.19 ± 11.91	83.81 ± 1.14	87.42 ± 0.56	96.39 ± 0.32	98.22 ± 0.60	94.31 ± 1.80	96.18 ± 1.35	98.48 ± 0.21	98.48 ± 0.20	**98.82 ± 0.03**

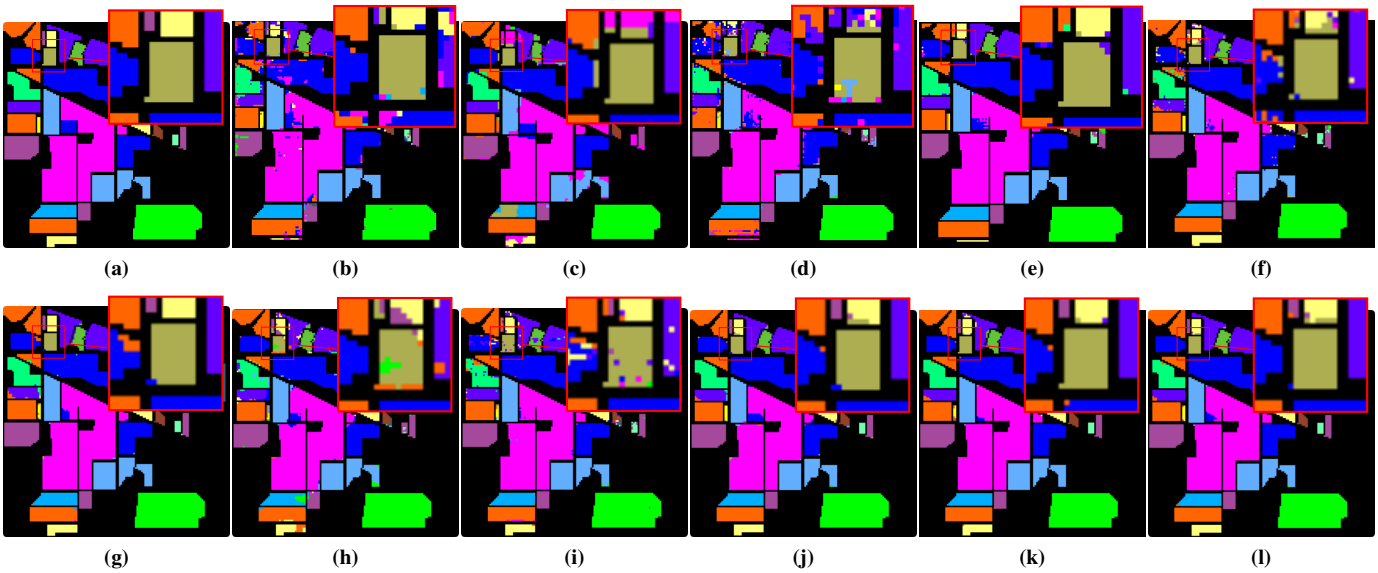


Fig. 5: Classification maps obtained by different methods on the Indian Pines dataset. (a) Ground Truth. (b) HiT. (c) SSFTT. (d) SS-TMNet. (e) DCTN. (f) SiT. (g) MambaHSI. (h) 3DSS-Mamba. (i) SS-Mamba. (j) IGroupSS-Mamba. (k) S2Mamba. (l) DTCMamba.

3) WHU-Hi-LongKou: On the large-scale WHU-Hi-LongKou dataset with an extremely scarce sampling ratio of 0.05%, DTCMamba demonstrates superior generalization capability, as reported in Table VI. Among the Transformer-based methods, SiT achieves an OA of 94.46%, showing clear improvement over earlier Transformer baselines. Among the Mamba-based competitors, IGroupSS-Mamba achieves the strongest competing OA of 95.29%, while MambaHSI and S2Mamba both obtain an OA of 95.21%. In contrast, DTCMamba achieves the highest OA of 95.84% and Kappa of 94.51%. More importantly, DTCMamba obtains the highest AA of 87.06%, outperforming IGroupSS-Mamba by 2.24%, which indicates stronger recognition ability for minority classes. This is especially important for WHU-Hi-LongKou, where the training samples are extremely limited and the class distribution is highly imbalanced. The superior AA suggests that the adaptive fusion mechanism can better balance spectral and spatial cues, reducing the bias toward majority classes and improving the discrimination of difficult categories.

Finally, Fig. 7 illustrates the classification maps on the WHU-Hi-LongKou dataset. It can be observed that 3DSS-Mamba and SSFTT exhibit scattered misclassification noise within large homogeneous crop fields. In contrast, the proposed DTCMamba demonstrates superior spatial continuity and sharper class boundaries, yielding a visual result highly consistent with the Ground Truth.

The visual improvements in Figs. 5–7 also provide qualitative evidence for the mitigation of boundary artifacts. Compared with competing methods, DTCMamba produces smoother homogeneous regions and clearer class boundaries.

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TABLE V: Comparison with State-of-the-Art methods on the Houston 2013 dataset (1% training samples). The best results are in bold.

Class	Transformers					Mamba					Ours
	HiT	SSFTT	SS-TMNet	DCTN	SiT	MambaHSI	3DSS-Mamba	SS-Mamba	IGroupSS-Mamba	S2Mamba	DTCMamba
1	86.93 ± 3.99	87.84 ± 7.92	90.48 ± 4.25	87.96 ± 4.00	94.81 ± 1.17	80.81 ± 1.06	66.24 ± 12.74	91.65 ± 2.39	96.22 ± 1.08	90.87 ± 3.74	**96.41 ± 1.37**
2	84.06 ± 9.58	82.42 ± 13.01	89.63 ± 4.83	94.80 ± 2.65	95.20 ± 1.39	92.54 ± 1.33	32.52 ± 7.76	96.02 ± 1.20	95.44 ± 1.38	94.83 ± 0.80	**98.37 ± 0.81**
3	86.60 ± 5.37	96.72 ± 5.54	96.24 ± 2.73	97.02 ± 2.65	96.55 ± 2.43	87.01 ± 0.55	88.32 ± 8.17	**99.68 ± 0.57**	92.81 ± 3.24	99.22 ± 0.92	94.58 ± 0.60
4	80.75 ± 7.11	75.83 ± 14.74	90.24 ± 2.21	**97.31 ± 0.60**	87.81 ± 2.90	85.63 ± 1.35	64.64 ±6.33	91.83 ± 3.14	88.81 ± 4.37	86.74 ± 3.52	96.48 ± 1.78
5	92.70 ± 2.71	96.89 ± 3.50	96.81 ± 0.49	97.16 ± 0.41	99.64 ± 0.76	**100.00 ± 0.00**	77.06 ± 4.42	99.19 ± 0.50	99.74 ± 0.54	**100.00 ± 0.00**	**100.00 ± 0.00**
6	33.91 ± 6.18	85.71 ± 5.50	69.94 ± 10.27	72.42 ± 20.69	85.40 ± 0.00	84.78 ± 1.22	17.83 ± 13.29	84.16 ± 4.53	85.09 ± 1.08	84.72 ± 2.05	**86.96 ± 0.22**
7	71.72 ± 4.60	76.38 ± 13.44	88.39 ± 1.89	93.34 ± 2.88	87.51 ± 2.89	85.47 ± 2.99	50.39 ± 11.69	89.29 ± 5.58	84.38 ± 2.20	88.70 ± 3.23	**94.60 ± 1.67**
8	62.16 ± 6.62	56.96 ± 16.33	71.87 ± 5.22	70.88 ± 5.32	66.57 ± 2.15	**72.40 ± 1.76**	51.43 ± 3.50	67.14 ± 2.96	68.52 ± 1.87	71.09 ± 0.87	67.45 ± 4.08
9	70.58 ± 7.33	66.15 ± 21.58	82.89 ± 5.25	87.81 ± 4.20	84.02 ± 5.15	84.68 ± 1.87	64.28 ± 2.21	81.21 ± 4.69	82.81 ± 7.82	87.83 ± 3.69	**92.36 ± 1.24**
10	78.94 ± 4.18	73.74 ± 25.26	82.62 ± 8.80	**98.80 ± 2.10**	93.81 ± 1.37	93.50 ± 1.21	74.72 ± 2.81	86.12 ± 7.23	93.53 ± 4.54	92.59 ± 0.63	96.02 ± 1.55
11	52.51 ± 3.32	64.81 ± 24.17	81.14 ± 8.02	82.26 ± 7.36	88.85 ± 2.92	93.16 ± 3.25	54.59 ± 9.92	87.05 ± 5.85	**98.66 ± 1.92**	97.89 ± 1.32	96.17 ± 3.31
12	67.48 ± 14.98	81.80 ± 20.94	82.79 ± 9.21	84.65 ± 5.36	89.70 ± 2.57	90.73 ± 3.14	61.74 ± 14.08	86.52 ± 5.16	**92.01 ± 1.70**	91.11 ± 3.16	90.89 ± 2.67
13	27.66 ± 4.97	33.23 ± 22.52	66.06 ± 10.27	71.36 ± 10.33	86.98 ± 7.15	87.93 ± 2.16	76.34 ± 9.91	49.48 ± 23.82	68.97 ± 6.69	67.84 ± 3.21	**89.35 ± 1.62**
14	51.98 ± 20.67	**100.00 ± 0.00**	92.78 ± 5.04	96.13 ± 5.89	**100.00 ± 0.00**	**100.00 ± 0.00**	97.17 ± 0.96	72.36 ± 14.09	99.86 ± 0.13	99.95 ± 0.11	**100.00 ± 0.00**
15	70.27 ± 8.10	84.87 ± 20.96	97.74 ± 5.06	99.11 ± 1.98	99.94 ± 0.14	**100.00 ± 0.00**	92.99 ± 5.39	99.82 ± 0.27	**100.00 ± 0.00**	**100.00 ± 0.00**	99.94 ± 0.08
OA (%)	72.14 ± 3.84	77.15 ± 4.28	85.97 ± 1.11	89.52 ± 0.81	89.82 ± 1.37	88.64 ± 0.67	63.17 ± 0.99	87.01 ± 1.15	90.08 ± 0.38	90.47 ± 0.68	**93.23 ± 0.16**
AA (%)	67.88 ± 3.60	77.56 ± 4.77	85.31 ± 1.77	88.73 ± 1.85	90.45 ± 1.20	89.24 ± 0.53	64.68 ± 0.68	85.43 ± 0.83	89.79 ± 0.44	90.23 ± 0.58	**93.31 ± 0.09**
$\kappa \times 100$	69.83 ± 4.17	75.28 ± 4.73	84.83 ± 1.20	88.68 ± 0.89	89.00 ± 1.48	89.24 ± 0.73	60.20 ± 1.05	85.94 ± 1.25	89.27 ± 0.41	89.70 ± 0.73	**92.68 ± 0.17**

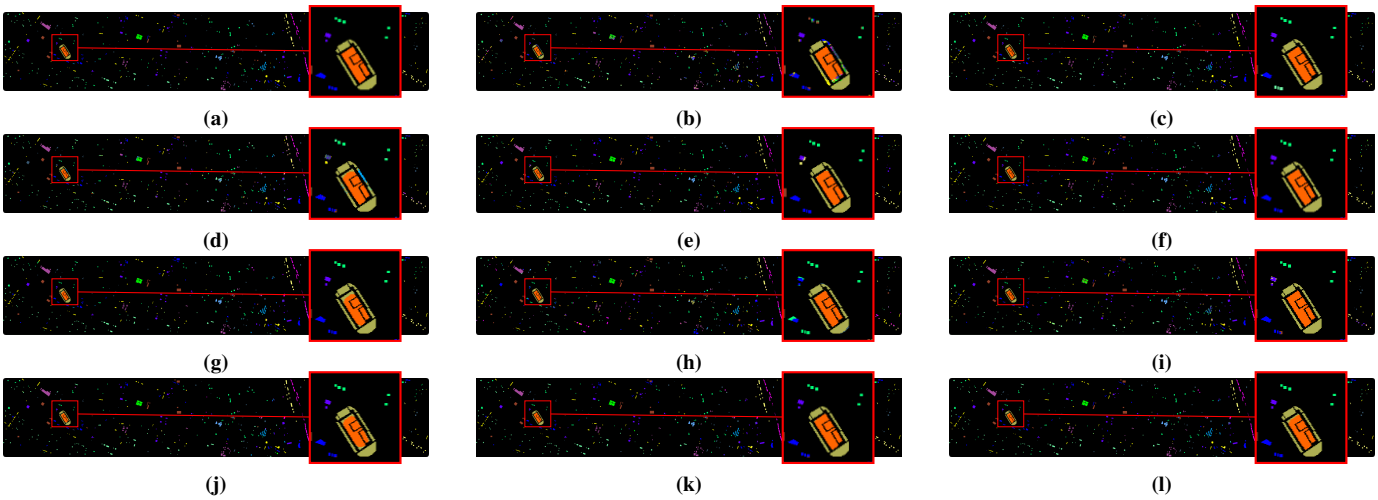


Fig. 6: Classification maps obtained by different methods on the Houston 2013 dataset. (a) Ground Truth. (b) HiT. (c) SSFTT. (d) SS-TMNet. (e) DCTN. (f) SiT. (g) MambaHSI. (h) 3DSS-Mamba. (i) SS-Mamba. (j) IGroupSS-Mamba. (k) S2Mamba. (l) DTCMamba.

TABLE VI: Comparison with State-of-the-Art methods on the WHU-Hi-LongKou dataset (0.05% training samples). The best results are in bold.

Class	Transformers					Mamba					Ours
	HiT	SSFTT	SS-TMNet	DCTN	SiT	MambaHSI	3DSS-Mamba	SS-Mamba	IGroupSS-Mamba	S2Mamba	DTCMamba
1	89.10 ± 0.46	57.00 ± 34.59	88.56 ± 1.25	88.39 ± 2.63	98.92 ± 0.55	98.88 ± 0.20	60.86 ± 13.97	94.88 ± 4.56	99.44 ± 0.13	99.25 ± 0.44	**99.46 ± 0.26**
2	56.75 ± 12.29	17.36 ± 23.43	64.28 ± 14.38	73.05 ± 7.72	76.21 ± 4.49	75.47 ± 1.97	12.70 ± 28.16	55.18 ± 18.40	80.17 ± 2.17	74.79 ± 1.65	**82.10 ± 4.47**
3	3.53 ± 3.31	24.83 ± 30.83	0.49 ± 0.60	6.09 ± 8.65	84.78 ± 2.39	86.77 ± 5.41	0.00 ± 0.00	25.80 ± 24.67	86.05 ± 0.70	59.46 ± 20.18	**88.20 ± 3.32**
4	86.42 ± 2.24	84.74 ± 8.65	89.90 ± 1.18	91.00 ± 0.80	99.38 ± 0.34	99.08 ± 0.15	88.81 ± 2.20	94.56 ± 3.85	98.84 ± 0.74	**99.70 ± 0.11**	98.94 ± 0.48
5	38.45 ± 10.52	13.49 ± 11.81	42.13 ± 18.70	51.32 ± 14.04	58.00 ± 3.47	59.12 ± 1.60	0.00 ± 0.00	43.88 ± 14.41	65.15 ± 3.78	59.58 ± 2.94	**73.30 ± 7.50**
6	88.07 ± 4.00	76.60 ± 26.65	86.29 ± 4.61	87.34 ± 5.66	94.31 ± 2.21	92.63 ± 1.70	49.94 ± 26.94	82.55 ± 10.89	**96.52 ± 1.40**	92.79 ± 2.86	95.90 ± 0.52
7	97.57 ± 0.01	98.15 ± 1.40	97.41 ± 0.38	97.54 ± 0.08	98.04 ± 0.61	99.00 ± 0.41	98.95 ± 1.33	**99.77 ± 0.31**	98.74 ± 0.43	99.90 ± 0.05	98.98 ± 0.50
8	62.99 ± 18.85	18.50 ± 32.33	53.15 ± 2.24	60.98 ± 21.91	54.53 ± 10.96	**71.51 ± 5.68**	0.61 ± 1.33	53.46 ± 21.68	63.31 ± 2.17	68.28 ± 6.42	61.43 ± 6.19
9	55.86 ± 17.32	45.93 ± 28.56	33.40 ± 9.08	44.92 ± 9.59	78.27 ± 5.48	78.81 ± 7.32	8.15 ± 15.67	51.86 ± 14.82	75.14 ± 4.23	77.87 ± 11.94	**85.24 ± 3.85**
OA (%)	85.61 ± 0.96	75.59 ± 4.60	85.86 ± 0.94	87.47 ± 1.25	94.46 ± 0.71	95.21 ± 0.22	73.80 ± 2.45	89.45 ± 2.98	95.29 ± 0.21	95.21 ± 0.52	**95.84 ± 0.17**
AA (%)	64.31 ± 5.12	48.51 ± 8.06	61.73 ± 1.91	66.74 ± 2.71	82.49 ± 1.35	84.59 ± 0.67	35.56 ± 3.54	66.88 ± 6.53	84.82 ± 0.50	81.29 ± 2.93	**87.06 ± 1.05**
$\kappa \times 100$	81.27 ± 1.32	66.91 ± 6.87	81.48 ± 1.23	83.58 ± 1.64	92.68 ± 0.94	93.66 ± 0.29	64.12 ± 3.57	85.94 ± 3.97	93.78 ± 0.28	93.63 ± 0.70	**94.51 ± 0.23**

This is closely related to the layered overlapping spiral scanning strategy, where the Center Overlapping Block acts as a spatial bridge between partitioned regions. By overlapping the central area and refining the target-pixel representation with an independent center encoder, the proposed spatial branch reduces discontinuities caused by patch partitioning and alleviates boundary-induced target contamination.

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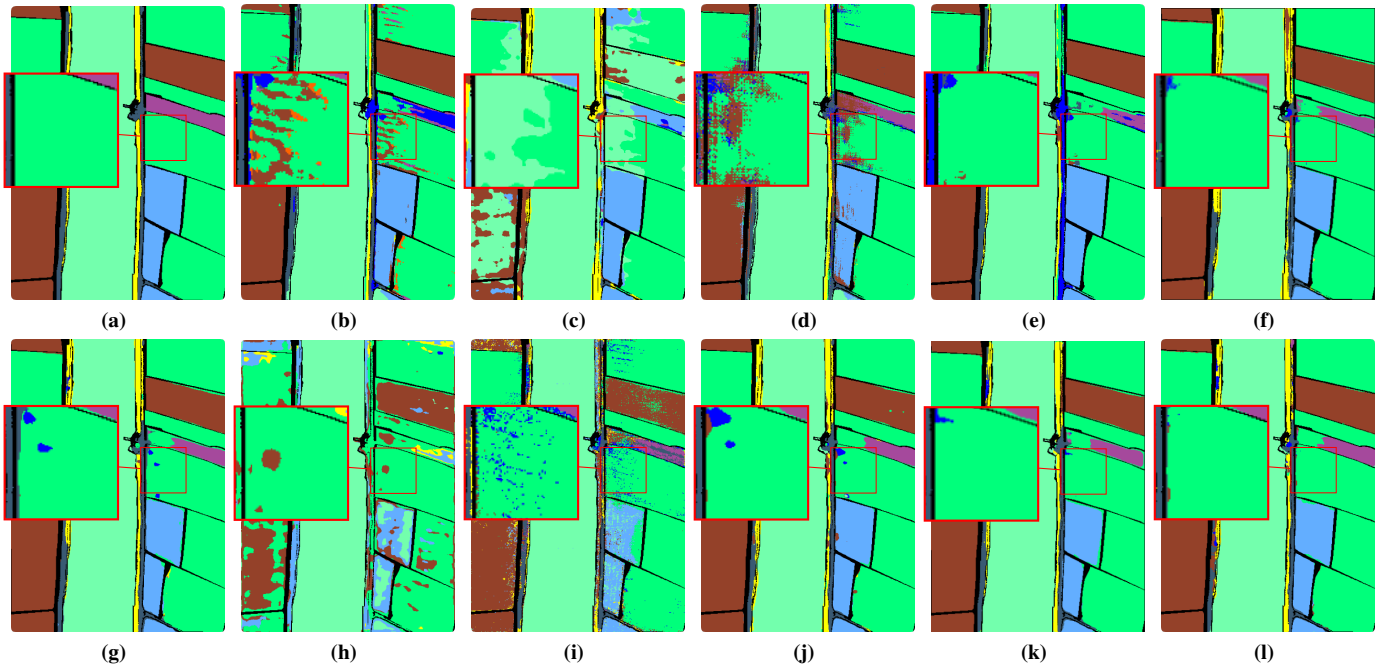


Fig. 7: Classification maps obtained by different methods on the WHU-Hi-LongKou dataset. (a) Ground Truth. (b) HiT. (c) SSFTT. (d) SS-TMNet. (e) DCTN. (f) SiT. (g) MambaHSI. (h) 3DSS-Mamba. (i) SS-Mamba. (j) IGroupSS-Mamba. (k) S2Mamba. (l) DTCMamba.

TABLE VII: Ablation study of different architecture components on the Indian Pines, Houston 2013, and WHU-Hi-LongKou datasets. The best results are in bold.

Architecture Variant	Components						Overall Accuracy (OA %)		
	Spatial	*Spectral*	*Residual*	*History*	*Center*	*Fusion*	Indian Pines	Houston 2013	WHU-Hi-LongKou
Spatial-Only	✓	×	✓	✓	×	✓	98.62	91.91	96.02
Spectral-Only	×	✓	✓	✓	×	✓	98.24	90.37	94.95
w/o Residual	✓	✓	×	✓	✓	✓	98.70	91.85	94.32
w/o History	✓	✓	✓	×	✓	✓	**98.86**	91.91	95.59
w/o Center Overlap	✓	✓	✓	✓	×	✓	98.67	91.64	95.66
w/o Adaptive Fusion	✓	✓	✓	✓	✓	×	98.76	92.71	95.83
DTCMamba (Full)	✓	✓	✓	✓	✓	✓	98.84	**92.92**	**96.25**

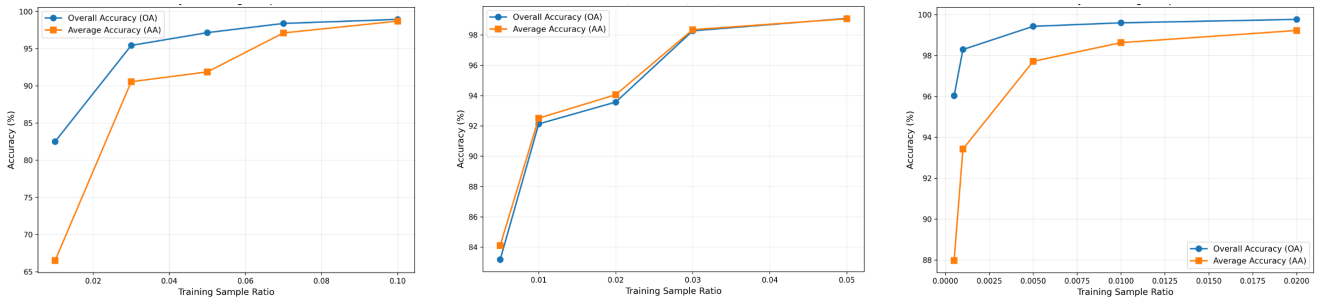


Fig. 8: Sensitivity analysis of training sample ratios on classification performance (OA and AA) across three datasets. (a) Indian Pines. (b) Houston 2013. (c) WHU-Hi-LongKou.

C. Ablation Study

We quantify the contribution of each proposed module in DTCMamba via ablation experiments on Indian Pines (IP), Houston 2013, and WHU-Hi-LongKou. Specifically, we evaluate three factors: (i) training-sample ratio (label efficiency), (ii) architectural components, and (iii) parallelization-driven

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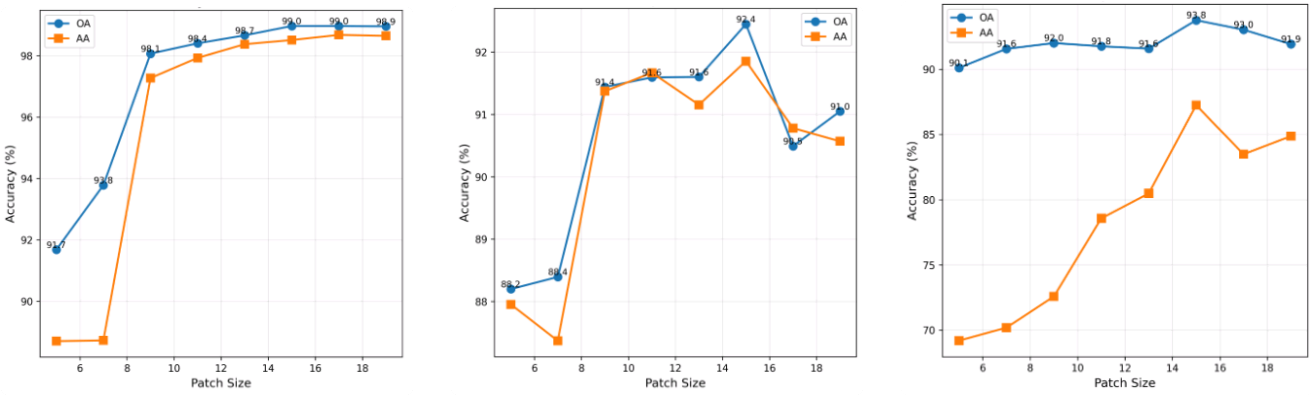


Fig. 9: Sensitivity analysis of patch size on classification performance (OA and AA) across three datasets. (a) Indian Pines. (b) Houston 2013. (c) WHU-Hi-LongKou.

efficiency. The ablated modules include History, cluster-based dual-branch spectral modeling, layered overlapping spiral scanning, and dual-branch adaptive fusion. All variants are trained under identical settings (optimizer, epochs, batch size, and data splits) to ensure fair comparison.

1) Impact of Training Sample Ratios : We assess label efficiency by varying the training-sample ratio and reporting both OA and AA across datasets, as illustrated in Fig. 8. On IP, DTCMamba achieves 82.51% OA with 1% training samples, and increasing the ratio to 10% improves OA to 98.94%, indicating that the model scales predictably with supervision. A consistent trend is observed on WHU-Hi-LongKou, where DTCMamba reaches 96.03% OA with only 0.05% training samples, supporting its effectiveness under severely limited labels.

2) Impact of Architecture Components : We ablate branch design and internal mechanisms using the variants in Table VII (Spatial-Only, Spectral-Only, w/o Residual, w/o History, w/o Center Overlap, w/o Adaptive Fusion , and Full). Table VII shows that the two-branch design yields the best overall performance across datasets. For example, on WHU-Hi-LongKou, the Full model reaches 96.25% OA, exceeding Spectral-Only (94.95%) and Spatial-Only (96.02%), indicating that spatial topology and spectral dependency provide complementary cues. We further isolate internal mechanisms: removing residual connections reduces OA (e.g., 96.25 $\rightarrow$ 94.32 on WHU), consistent with residuals improving optimization stability in deep SSM blocks. While w/o History remains competitive on IP, the Full model maintains the best or comparable OA across datasets, suggesting that History is more beneficial under harder long-sequence spectral modeling. Specifically, the 0.02% OA difference between w/o History and the Full model on Indian Pines is very small and can be regarded as comparable performance, while the Full model achieves clear improvements on Houston 2013 and WHU-Hi-LongKou.

In addition, the w/o Center Overlap variant obtains lower OA than the full model on all three datasets, indicating that the Center Overlapping Block helps preserve the target-pixel representation and reduce background interference. Finally, replacing the adaptive fusion gate with a simple feature summation (w/o Adaptive Fusion) decreases the OA across datasets (e.g., 96.25% $\rightarrow$ 95.83% on WHU), confirming that dynamically learning pixel-wise mixing weights is superior to static fusion for integrating spectral-spatial features.

The ablation results further explain why the history-enhanced mechanism is useful for long spectral modeling. In long spectral sequences, discriminative information from distant bands may be gradually weakened during recurrent state propagation, which can make the model rely more on nearby or recently scanned bands. The History module introduces a gated second-order historical state, providing an additional propagation path for previous spectral states. Although the improvement is dataset-dependent, the ablation results show that the proposed history-enhanced mechanism helps stabilize long-range spectral modeling, especially on more challenging datasets with complex spectral-spatial distributions.

3) Impact of Parallelization Strategy : We evaluate the inference-latency benefit of Parallel TCSSD by comparing Parallel vs. Sequential execution under the same model parameters. Due to the long spectral length, sequential SSM recurrence incurs high latency; Parallel TCSSD reduces the effective sequential depth via chunk-wise parallel execution. Table VIII reports the average inference time per epoch on three datasets. Parallel execution consistently reduces latency—for example, from 414.5 ms to 306.2 ms on IP ($\approx$26%) and from 311.4 ms to 256.6 ms on WHU ($\approx$18%). These results indicate that Parallel TCSSD improves latency without changing the parameterization, and the gain primarily comes from execution parallelism.

D. Parameter Analysis

In this section, we conduct a sensitivity analysis of DTC-Mamba to three key hyperparameters: The input patch size (P), Parallel Chunk Size (C), and the Spectral Group Number (K).

1) Analysis of Patch Size (P) : The input patch size P determines the model's effective spatial receptive field and governs the trade-off between capturing sufficient context and

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TABLE VIII: Inference time comparison between Sequential (Standard) and Parallel (Ours) processing modes. The results are averaged over 5 test epochs.

Dataset	Mode	Inference Time (ms)
Indian Pines	Sequential	414.5
	Parallel (Ours)	**306.2**
Houston 2013	Sequential	371.3
	Parallel (Ours)	**334.5**
WHU-Hi-LongKou	Sequential	311.4
	Parallel (Ours)	**256.6**

avoiding cross-class contamination from heterogeneous neighbors. We evaluated P over 5,7,...,19 on all three datasets, and the resulting OA/AA trends are summarized in Fig. 9. Indian Pines: performance increases monotonically and then saturates; the coarse resolution (20 m) and homogeneous fields favor larger windows that capture continuous field structures with limited boundary-induced noise, with saturation at $P = 15$. Houston 2013: Fig. 9(b) exhibits an inverted-U trend—moderate P provides shape cues, whereas overly large P introduces mixed pixels from adjacent classes, diluting class-specific spatial cues. WHU-Hi-LongKou: performance saturates early at $P = 15$ due to the extremely high resolution (0.463 m), where small windows already contain abundant spatial details and additional context becomes less beneficial. Considering accuracy–efficiency trade-offs across datasets, we set $P = 15$ as the default.

2) Analysis of Parallel Chunk Size (C) : The chunk size C in Parallel-TCSSD is a key hyperparameter that trades off hardware parallelism against within-chunk sequential depth. Larger C increases the number of recurrent steps per chunk, which may improve short-range dependency modeling but incurs higher sequential latency, whereas smaller C increases parallelism but introduces chunk/state-management overhead. As shown in Table IX, decreasing C from 32 to 8 reduces inference time substantially across datasets (e.g., 335→277 ms on WHU-Hi-LongKou), consistent with a reduced effective sequential depth. Further decreasing C to 2/4 yields diminishing returns and can even increase latency on WHU-Hi-LongKou, likely due to state-management overhead, while accuracy remains largely stable with minor fluctuations. Prioritizing real-time deployment, we choose $C = 8$ as the default (e.g., 352.72 ms with 98.78% accuracy on Indian Pines), and fix it in subsequent experiments since larger C brings only marginal accuracy gains (e.g., 96.45% on WHU) relative to the latency/stability advantages.

3) Analysis of the Spectral Group Number (K) : The spectral group number K controls the granularity of K-Means-based spectral decoupling. A small K leads to lower computational cost but may be insufficient to separate redundant and informative spectral bands, while a large K provides finer spectral grouping but introduces additional model parameters and computational overhead. Therefore, we evaluate the influence of K on Indian Pines, Houston 2013, and WHU-Hi-LongKou by varying K from 1 to 8. As shown in Table X, increasing K generally improves the spectral decoupling ability, but the model complexity and inference time also increase accordingly. On Indian Pines, $K = 4$ achieves the best OA of 98.88%, while further increasing K brings no additional improvement. On Houston 2013 and WHU-Hi-LongKou, $K = 8$ obtains slightly higher OA than $K = 4$, but this improvement is accompanied by a clear increase in parameters, FLOPs, and inference time. For example, compared with $K = 4$, $K = 8$ increases the parameters from 1.25M to 2.47M, the FLOPs from 0.12G to 0.23G, and the inference time from 221.44 ms to 340.48 ms on Houston 2013, while the OA only increases from 90.19% to 90.46%. Therefore, considering both classification accuracy and computational efficiency, $K = 4$ is adopted as the default setting in our experiments. For datasets with different spectral characteristics or deployment constraints, K can be selected according to validation performance and computational resources.

E. Feature Visualization

To qualitatively assess feature discriminability, we use t-SNE to visualize the learned feature embeddings of DCTN [31], IGroupSS-Mamba [20], and DTCMamba on Indian Pines (Fig. 10). DCTN [31] (Fig. 10(a)) shows dispersed clusters with substantial overlap among spectrally similar vegetation classes, notably Soybean and Corn . IGroupSS-Mamba [20] (Fig. 10(b)) improves intra-class compactness, yet ambiguous boundaries remain in these confusable categories. In comparison, DTCMamba (Fig. 10(c)) yields more compact intra-class clusters and clearer inter-class separation, forming dense clusters even for highly entangled Soybean and Corn classes. Overall, these observations suggest that DTCMamba learns embeddings with improved class separability; this is consistent with the intended roles of History-Enhanced Parallel TCSSD and the cluster-based dual-branch spectral strategy in mitigating long-sequence state decay and spectral redundancy.

F. Complexity Analysis

Table XI reports the computational complexity of different Transformer- and Mamba-based methods on the Houston 2013 dataset, including parameters, FLOPs, inference time, training time per epoch, and GPU memory usage. Among the Transformer-based methods, HiT, SS-TMNet, and DCTN contain relatively large numbers of parameters, while SiT has a compact model size of 0.22M parameters but requires 4.77G FLOPs due to its scale-interaction modeling. Among the Mamba-based methods, MambaHSI and IGroupSS-Mamba are lightweight in terms of parameters and FLOPs, whereas S2Mamba requires higher memory usage because of its spatial-spectral state modeling design. Compared with these baselines, DTCMamba has a moderate model size with 1.80M parameters and 0.15G FLOPs. Although its inference time is not the lowest, DTCMamba achieves the best classification performance on Houston 2013, as shown in Table V. These results indicate that DTCMamba provides a favorable trade-off between classification accuracy and computational cost, benefiting from the cluster-based spectral decoupling, parallel TCSSD modeling, and layered overlapping spatial scanning strategy.

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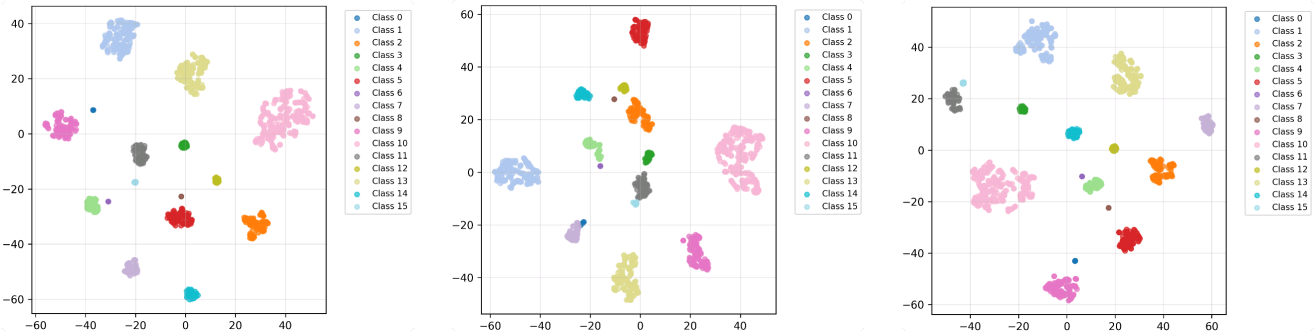
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TABLE IX: Parameter sensitivity analysis of Parallel Chunk Size (C) on Indian Pines, Houston 2013, and WHU-Hi-LongKou datasets. Inference Time is the primary metric for efficiency.

Chunk Size (C)	Indian Pines				Houston 2013				WHU-Hi-LongKou			
	Time (ms) ↓	OA (%)	AA (%)	$\kappa \times 100$	Time (ms) ↓	OA (%)	AA (%)	$\kappa \times 100$	Time (ms) ↓	OA (%)	AA (%)	$\kappa \times 100$
2	875.94	98.83	98.07	98.66	308.38	93.10	93.15	92.54	349.83	95.51	85.96	94.06
4	635.05	98.92	97.41	98.76	275.83	**93.47**	**93.01**	**92.94**	**275.97**	95.49	87.22	94.06
8	**352.72**	98.78	97.32	98.60	272.18	93.05	92.69	92.49	277.14	95.93	86.84	94.63
16	365.26	**99.00**	**98.92**	**98.86**	**266.59**	91.97	92.33	91.32	277.32	96.13	87.49	94.89
32	359.31	98.88	98.47	98.73	300.13	92.78	92.92	92.19	335.30	**96.45**	**88.74**	**95.31**

TABLE X: Parameter sensitivity analysis of the spectral group number (K) on Indian Pines, Houston 2013, and WHU-Hi-LongKou datasets. Parameters, FLOPs, and inference time are reported to evaluate the accuracy-efficiency trade-off.

Group Number (K)	Indian Pines				Houston 2013				WHU-Hi-LongKou			
	OA (%)	Params (M)	FLOPs (G)	Time (ms) ↓	OA (%)	Params (M)	FLOPs (G)	Time (ms) ↓	OA (%)	Params (M)	FLOPs (G)	Time (ms) ↓
1	97.98	**0.34**	**0.03**	**44.16**	88.84	**0.34**	**0.03**	**60.80**	93.79	**0.34**	**0.03**	**41.60**
2	98.34	0.64	0.06	96.64	89.67	0.64	0.06	120.32	93.75	0.64	0.06	89.60
4	**98.88**	1.25	0.12	158.08	90.19	1.25	0.12	221.44	94.86	1.25	0.12	158.08
6	98.49	1.86	0.18	232.96	90.38	1.86	0.18	266.88	94.68	1.86	0.18	223.36
8	98.47	2.47	0.23	301.44	**90.46**	2.47	0.23	340.48	**95.25**	2.47	0.23	289.28



(a) (b) (c)

Fig. 10: t-SNE feature visualization of different methods on the Indian Pines dataset. (a) DCTN[31] . (b) IGroupSS-Mamba [20] . (c) The proposed DTCMamba.

TABLE XI: Complexity analysis of different models on the Houston 2013 dataset. The metrics include Parameters (Params), Floating Point Operations (FLOPs), Inference Time (InfTime), Training Time per Epoch (TrainTime), and GPU Memory Usage.

Model	Params (M)	FLOPs (G)	InfTime (ms)	TrainTime (s/ep)	Memory (MB)
HiT	46.13	0.89	213,069.75	0.27	1,057.94
SSFTT	0.15	0.02	0.60	0.02	18.49
SS-TMNet	68.67	1.74	570,678.90	0.53	3,112.78
DCTN	39.09	1.07	357,456.89	0.29	2,123.66
SiT	0.22	4.77	0.12	0.05	1,069.71
MambaHSI	0.05	0.01	2.16	0.07	18.89
3DSS-Mamba	0.01	0.03	0.71	0.36	36.49
SS-Mamba	0.95	0.21	101.54	31.08	37.02
IGroupSS-Mamba	0.06	0.01	30.00	0.60	18.42
S2Mamba	0.07	1.04	0.86	9.99	4,808.84
DTCMamba (Ours)	1.80	0.15	225.37	1.99	32.54

VI. CONCLUSION

In this paper, we present DTCMamba, a dual-branch token-chunking Mamba-based framework for hyperspectral image classification. DTCMamba integrates (i) a history-enhanced parallel TCSSD block for long spectral modeling, (ii) a cluster-based spectral decoupling strategy to reduce inter-band redundancy, and (iii) layered overlapping spiral scanning to better preserve 2-D topology and mitigate boundary artifacts. A pixel-wise adaptive fusion gate further balances spectral and spatial cues. Experiments on Indian Pines, Houston 2013, and WHU-Hi-LongKou show that DTCMamba outperforms recent Transformer- and Mamba-based baselines, with larger gains under severely limited labels (e.g., 0.05% training samples). Future work will extend DTCMamba to other remote-

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sensing tasks and explore efficient deployment (e.g., pruning/quantization) under resource constraints.

VII. ACKNOWLEDGMENT

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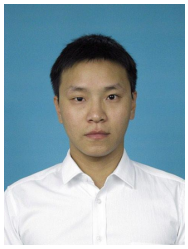
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